INCH-POUND

MIL-DTL-83503/23C
21 July 2004
SUPERSEDING
MIL-C-83503/23B
17 June 1996

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL PRINTED WIRING BOARD SOLDER TRANSITION FOR TERMINATING ROUND CONDUCTORS ON .050 INCH CENTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83503.

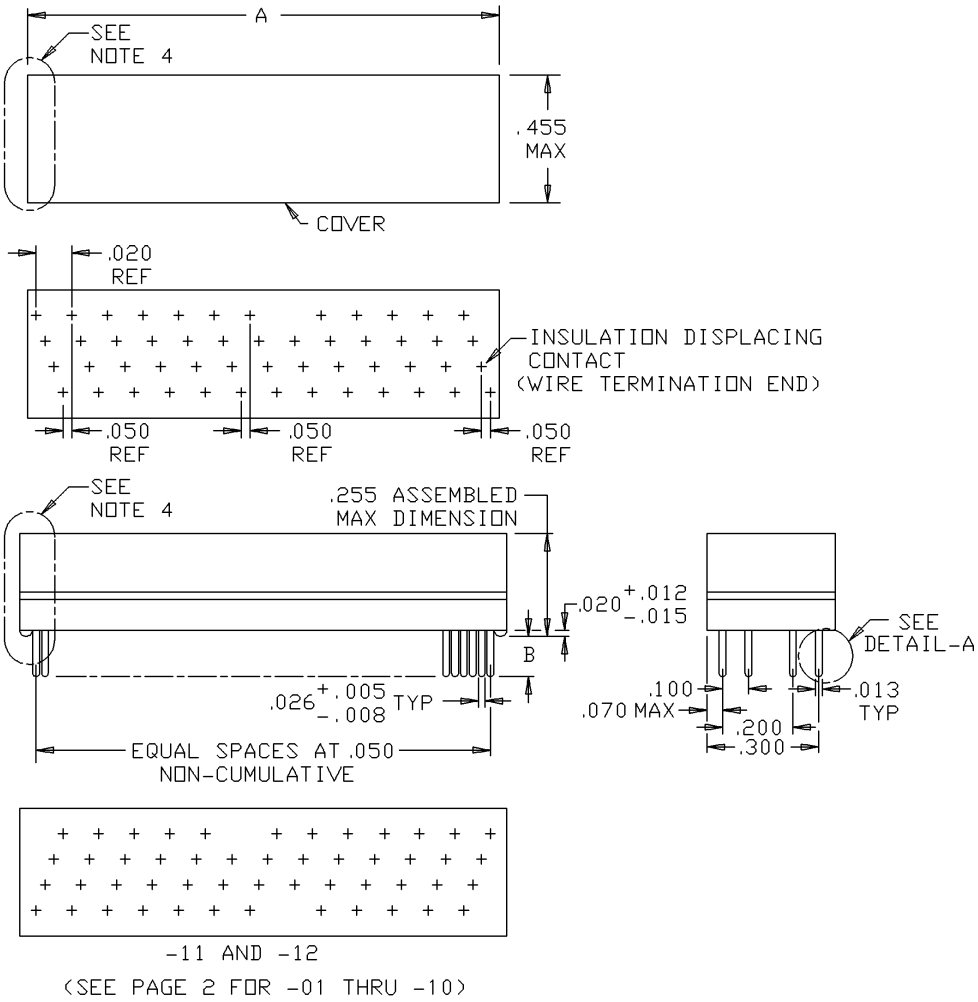


FIGURE 1. Dimensions and configuration.

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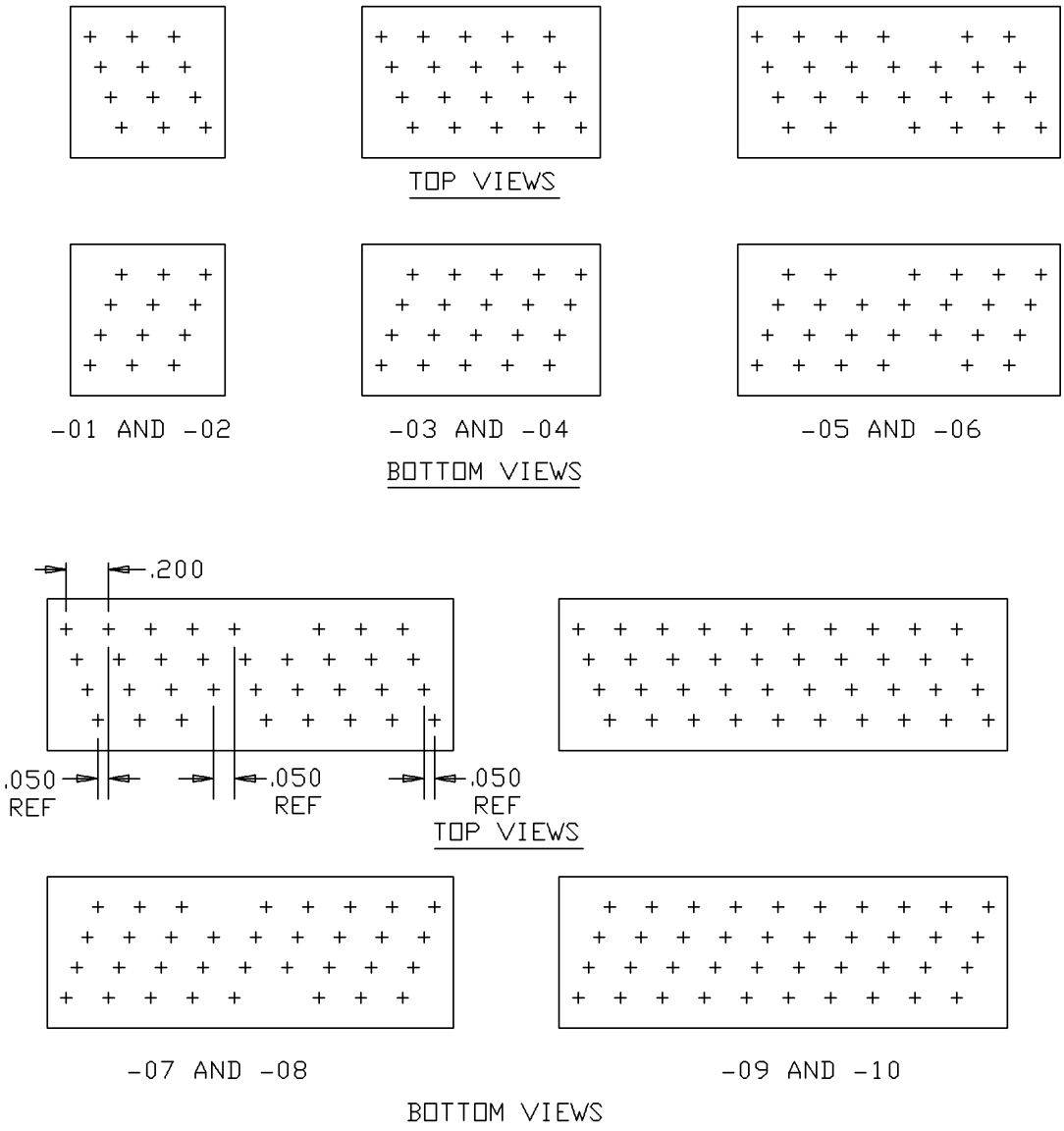
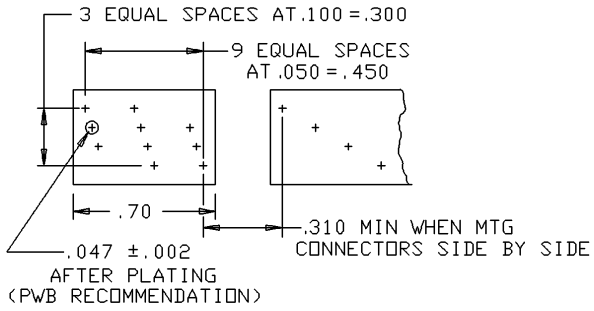
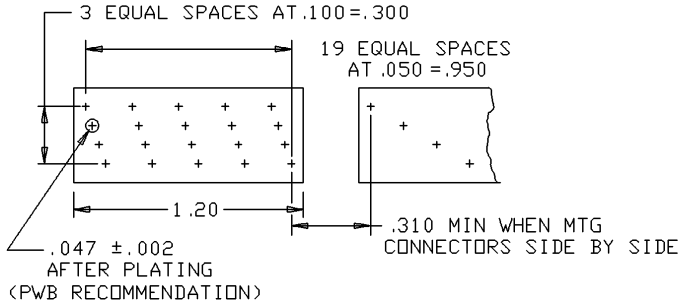


FIGURE 1. Dimensions and configuration - Continued.

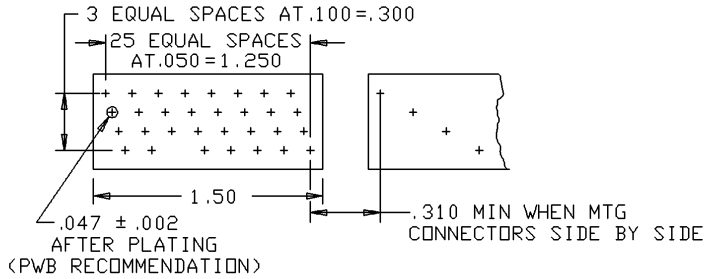
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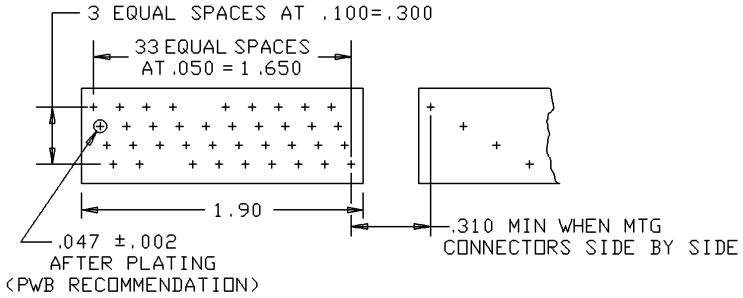
M83503/23 -01 AND -02



M83503/23 -03 AND -04



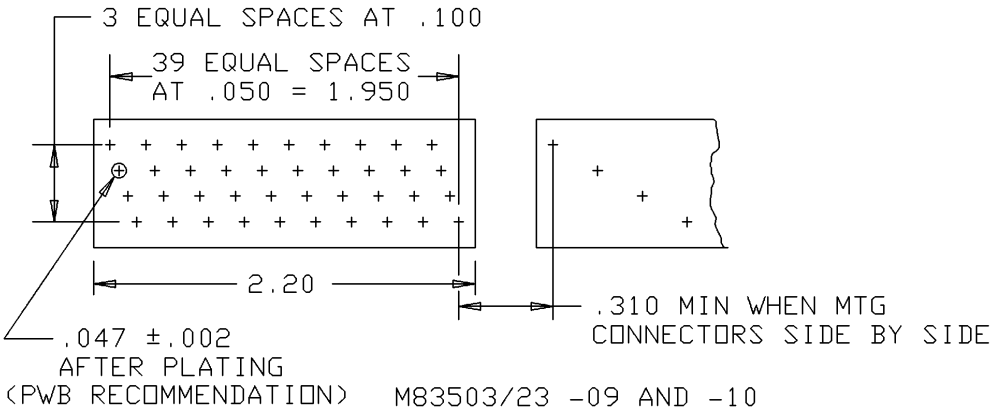
M83503/23 -05 AND -06

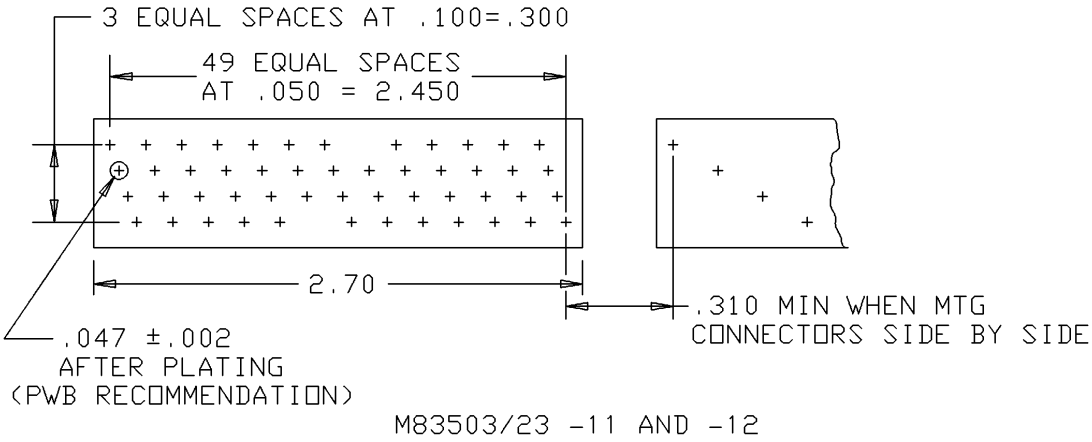


M83503/23 -07 AND -08

FIGURE 1. Dimensions and configuration - Continued.

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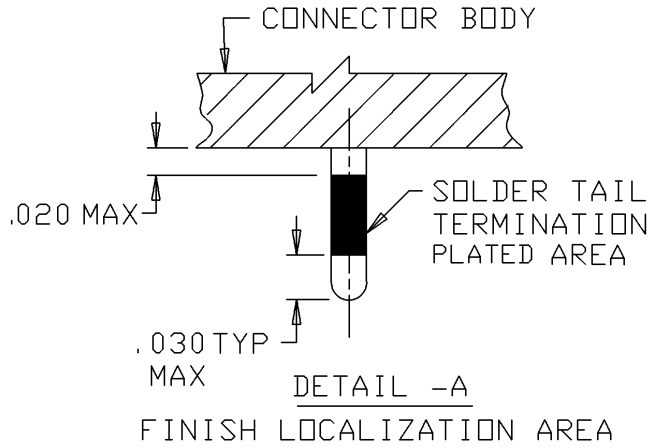


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.047	1.19	.950	24.13
.005	0.13	.050	1.27	1.20	30.5
.008	0.20	.070	1.78	1.250	31.75
.012	0.30	.100	2.54	1.50	38.1
.013	0.33	.200	5.08	1.650	41.91
.015	0.38	.255	6.48	1.90	48.3
.020	0.51	.300	7.62	1.950	49.53
.026	0.66	.310	7.87	2.20	55.9
.030	0.76	.450	11.43	2.450	62.23
		.455	11.56	2.70	68.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .02 inch (0.5 mm) for two place decimals.
4. The method used to connect cover to body is optional, but shall meet the requirements of MIL-DTL-83503 and this specification sheet.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	Number of positions	Dimension	
		A inch (mm) max	B inch (mm) max
-01	10	.72 (18.3)	.100 (2.54)
-02	10	.72 (18.3)	.156 (3.96)
-03	20	1.22 (31.0)	.100 (2.54)
-04	20	1.22 (31.0)	.156 (3.96)
-05	26	1.52 (38.6)	.100 (2.54)
-06	26	1.52 (38.6)	.156 (3.96)
-07	34	1.92 (48.3)	.100 (2.54)
-08	34	1.92 (48.3)	.156 (3.96)
-09	40	2.22 (56.4)	.100 (2.54)
-10	40	2.22 (56.4)	.156 (3.96)
-11	50	2.72 (69.1)	.100 (2.54)
-12	50	2.72 (69.1)	.156 (3.96)

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° to +120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Contacts shall be phosphor bronze or beryllium copper. Phosphor bronze shall be in accordance with ASTM B139 spring temper or brass in accordance with CDA 260 spring temper. Beryllium copper shall be in accordance with ASTM-B196 or ASTM B197 alloy UNS No. C17200 spring temper.

Plating: Tin-lead in the termination area in accordance with MIL-DTL-83503.

Housing: Glass field polyester in accordance with MIL-M-24519, type GPT-15F, GPT-20F, GPT-30F.

Wire accommodation: 28 AWG stranded on .050 (1.27 mm) inch spacing between conductor centerlines.

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Voltage rating: 300 volts.

Test wire size: 28 AWG stranded.

Preconditioning: Not applicable.

Withstanding voltage:

Sea level: 500 V rms, minimum.

Altitude: 200 V rms, minimum.

Cable retention: Not applicable.

Contact resistance: 50 milliohms, maximum.

Mating and unmating force: Not applicable.

Durability: Not applicable.

Interface:

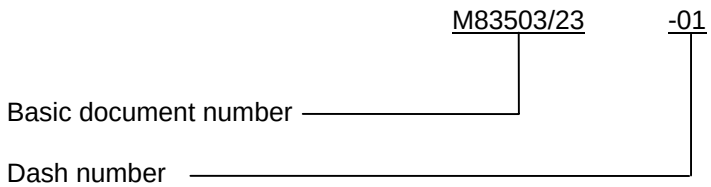
Round conductor flat cable: .050 inch (1.27 mm) centers to .062 inch (1.57 mm).

Printed wiring boards (thickness): .093 inch (2.36 mm) to .125 inch (3.18 mm).

Marking:

Part or Identifying Number (PIN): The PIN shall consist of the letter “M”, followed by the basic specification number, “/”, the specification sheet number, “-” and the dash number from table I.

Example:



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Referenced documents: In addition to MIL-DTL-83503, this document references the following:

MIL-M-24519
ASTM B139
ASTM B196
ASTM B197
CDA 260

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4601-006)

Review activities:
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

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